Task

Task:Create the K8s EKS,further you have to do the deployment of the Nginx application and access the application outside the cluster.

1.installing kubectl,eksctl and aws cli.

```
ubuntu@ip-172-31-46-170:~$ curl -LO https://dl.k8s.io/release/$(curl -L -s https
://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl

chmod +x kubectl
sudo mv kubectl /usr/local/bin/
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 55.8M  100 55.8M    0     0   235M      0 --:--:-- --:--:-- --:--:--  234M
ubuntu@ip-172-31-46-170:~$ kubectl version --client
Client Version: v1.35.1
Kustomize Version: v5.7.1
ubuntu@ip-172-31-46-170:~$
```

```
ubuntu@ip-172-31-46-170:~$ curl --silent --location \
"https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s
)_amd64.tar.gz" \
| tar xz -C /tmp

sudo mv /tmp/eksctl /usr/local/bin
ubuntu@ip-172-31-46-170:~$ eksctl version
0.223.0
ubuntu@ip-172-31-46-170:~$ |
```

```
ubuntu@ip-172-31-46-170:~$ /usr/local/bin/aws --version
aws-cli/2.33.23 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-46-170:~$
```

2. Create EKS Cluster

```
ubuntu@ip-172-31-46-170:~$ eksctl create cluster   --name my-eks-cluster   -
-region eu-north-1   --nodes 1   --node-type t3.small   --managed
2026-02-17 09:09:08 [ℹ]  eksctl version 0.223.0
2026-02-17 09:09:08 [ℹ]  using region eu-north-1
2026-02-17 09:09:08 [ℹ]  setting availability zones to [eu-north-1b eu-north
-1c eu-north-1a]
2026-02-17 09:09:08 [ℹ]  subnets for eu-north-1b - public:192.168.0.0/19 pri
vate:192.168.96.0/19
2026-02-17 09:09:08 [ℹ]  subnets for eu-north-1c - public:192.168.32.0/19 pr
ivate:192.168.128.0/19
2026-02-17 09:09:08 [ℹ]  subnets for eu-north-1a - public:192.168.64.0/19 pr
ivate:192.168.160.0/19
2026-02-17 09:09:08 [ℹ]  nodegroup "ng-108955de" will use "" [AmazonLinux202
3/1.34]
```

```
2026-02-17 09:18:12 [ℹ]  creating addon: vpc-cni
2026-02-17 09:18:12 [ℹ]  successfully created addon: vpc-cni
2026-02-17 09:20:12 [ℹ]  building managed nodegroup stack "eksctl-my-eks-cluster-nodegroup-ng-108955de"
2026-02-17 09:20:13 [ℹ]  deploying stack "eksctl-my-eks-cluster-nodegroup-ng-108955de"
2026-02-17 09:20:13 [ℹ]  waiting for CloudFormation stack "eksctl-my-eks-cluster-nodegroup-ng-108955de"
2026-02-17 09:20:43 [ℹ]  waiting for CloudFormation stack "eksctl-my-eks-clu
ster-nodegroup-ng-108955de"
2026-02-17 09:21:38 [ℹ]  waiting for CloudFormation stack "eksctl-my-eks-clu
ster-nodegroup-ng-108955de"
2026-02-17 09:22:57 [ℹ]  waiting for CloudFormation stack "eksctl-my-eks-clu
ster-nodegroup-ng-108955de"
2026-02-17 09:22:57 [ℹ]  waiting for the control plane to become ready
2026-02-17 09:22:58 [✔]  saved kubeconfig as "/home/ubuntu/.kube/config"
2026-02-17 09:22:58 [ℹ]  no tasks
2026-02-17 09:22:58 [✔]  all EKS cluster resources for "my-eks-cluster" have
 been created
2026-02-17 09:22:58 [ℹ]  nodegroup "ng-108955de" has 1 node(s)
2026-02-17 09:22:58 [ℹ]  node "ip-192-168-43-111.eu-north-1.compute.internal
" is ready
2026-02-17 09:22:58 [ℹ]  waiting for at least 1 node(s) to become ready in "
ng-108955de"
2026-02-17 09:22:58 [ℹ]  nodegroup "ng-108955de" has 1 node(s)
2026-02-17 09:22:58 [ℹ]  node "ip-192-168-43-111.eu-north-1.compute.internal
" is ready
2026-02-17 09:22:58 [✔]  created 1 managed nodegroup(s) in cluster "my-eks-c
luster"
2026-02-17 09:22:58 [ℹ]  creating addon: metrics-server
2026-02-17 09:22:58 [ℹ]  successfully created addon: metrics-server
2026-02-17 09:22:59 [ℹ]  kubectl command should work with "/home/ubuntu/.kub
e/config", try 'kubectl get nodes'
2026-02-17 09:22:59 [✔]  EKS cluster "my-eks-cluster" in "eu-north-1" region
 is ready
```

```
ubuntu@ip-172-31-46-170:~$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-192-168-43-111.eu-north-1.compute.internal   Ready    <none>   7m51s   v1.34.3-eks-70ce843
ubuntu@ip-172-31-46-170:~$
```

3. Deploy Nginx Application

```
ubuntu@ip-172-31-46-170:~$ kubectl create deployment nginx-deployment --image=nginx
deployment.apps/nginx-deployment created
ubuntu@ip-172-31-46-170:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7457467ffd-sx7t4   1/1     Running   0          31s
ubuntu@ip-172-31-46-170:~$
```

4. Expose Application Outside Cluster

```
ubuntu@ip-172-31-46-170:~$ kubectl expose deployment nginx-deployment \
  --type=LoadBalancer \
  --port=80
service/nginx-deployment exposed
ubuntu@ip-172-31-46-170:~$
```

5. Get external URL

```
ubuntu@ip-172-31-46-170:~$ kubectl get svc nginx-deployment
NAME               TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)        AGE
nginx-deployment   LoadBalancer   10.100.101.15   a5a10cdfc44b8461984a34b604037958-1333727784.eu-north-1.elb.amazonaws.com   80:31559/TCP   71s
ubuntu@ip-172-31-46-170:~$
```

6. Open this in browser

Not secure a5a10cdfc44b8461984a34b604037958-1333727784.eu-north-1.elb.amazonaws.com

Maps

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

7. Deploy Nginx Application with yaml file

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
~
~
```

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: LoadBalancer
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
~
~
~
~
```

```
ubuntu@ip-172-31-46-170:~$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
```

```
deployment.apps/nginx-deployment unchanged
ubuntu@ip-172-31-46-170:~$ kubectl apply -f service.yaml
service/nginx-service created
ubuntu@ip-172-31-46-170:~$
```

8. verifying the deployment and service.

```
service/nginx-service created
ubuntu@ip-172-31-46-170:~$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           4m34s
ubuntu@ip-172-31-46-170:~$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP               NODE                                           NOMINATED NODE   READINESS GATES
nginx-deployment-75fdcbbc74-6qtg9   1/1     Running   0          4m48s   192.168.34.226   ip-192-168-43-111.eu-north-1.compute.internal   <none>           <none>
nginx-deployment-75fdcbbc74-75bwk   1/1     Running   0          4m48s   192.168.57.32    ip-192-168-43-111.eu-north-1.compute.internal   <none>           <none>
ubuntu@ip-172-31-46-170:~$ kubectl get svc
NAME              TYPE           CLUSTER-IP      EXTERNAL-IP                                                                   PORT(S)        AGE
kubernetes        ClusterIP      10.100.0.1      <none>                                                                        443/TCP        30m
nginx-deployment  LoadBalancer   10.100.101.15   a5a10cdfc44b8461984a34b604037958-1333727784.eu-north-1.elb.amazonaws.com   80:31559/TCP   14m
nginx-service     LoadBalancer   10.100.134.84   a22a5cd34bccc46ca98beec85eed4ef2-1457490393.eu-north-1.elb.amazonaws.com   80:32086/TCP   3m18s
ubuntu@ip-172-31-46-170:~$
```

```
ubuntu@ip-172-31-46-170:~$ kubectl get svc
NAME              TYPE           CLUSTER-IP       EXTERNAL-IP                                                                   PORT(S)        AGE
kubernetes        ClusterIP      10.100.0.1       <none>                                                                        443/TCP        53m
nginx-service     LoadBalancer   10.100.109.126   a7f1facd6fe40446482567794ab918fd-1428899793.eu-north-1.elb.amazonaws.com   80:32049/TCP   9m33s
ubuntu@ip-172-31-46-170:~$
```

9. troubleshooting

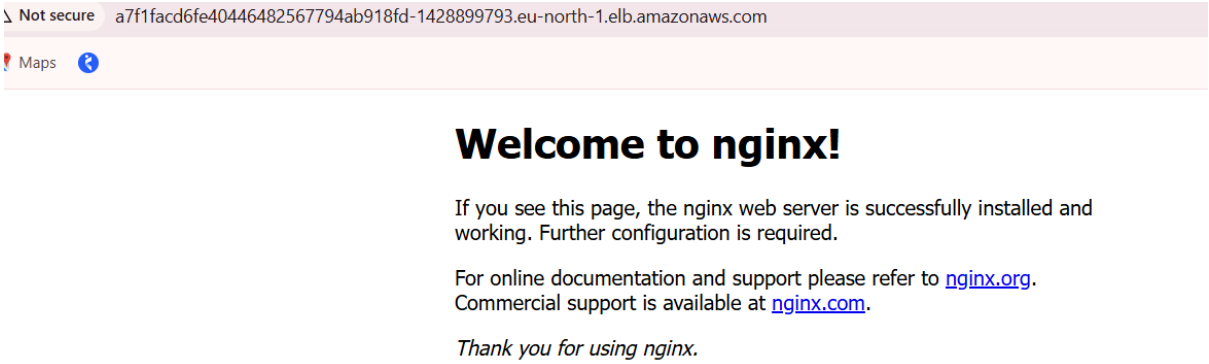
```
ubuntu@ip-172-31-46-170:~$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP               NODE                                           NOMINATED NODE   READINESS GATES
nginx-deployment-75fdcbbc74-6qtg9   1/1     Running   0          10m   192.168.34.226   ip-192-168-43-111.eu-north-1.compute.internal   <none>           <none>
nginx-deployment-75fdcbbc74-75bwk   1/1     Running   0          10m   192.168.57.32    ip-192-168-43-111.eu-north-1.compute.internal   <none>           <none>
ubuntu@ip-172-31-46-170:~$ kubectl get endpoints nginx-deployment
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME               ENDPOINTS   AGE
nginx-deployment   <none>      20m
ubuntu@ip-172-31-46-170:~$ kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
nginx-deployment-75fdcbbc74-6qtg9   1/1     Running   0          14m   app=nginx,pod-template-hash=75fdcbbc74,topology.kubernetes.io/region=eu-north-1,topology.kubernetes.io/zone=eu-north-1c
nginx-deployment-75fdcbbc74-75bwk   1/1     Running   0          14m   app=nginx,pod-template-hash=75fdcbbc74,topology.kubernetes.io/region=eu-north-1,topology.kubernetes.io/zone=eu-north-1c
ubuntu@ip-172-31-46-170:~$ kubectl describe svc nginx-deployment
Name:                     nginx-deployment
Namespace:                default
Labels:                   app=nginx-deployment
Annotations:              <none>
Selector:                 app=nginx-deployment
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.101.15
IPs:                      10.100.101.15
LoadBalancer Ingress:     a5a10cdfc44b8461984a34b604037958-1333727784.eu-north-1.elb.amazonaws.com
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  31559/TCP
Endpoints:
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:
  Type    Reason                Age   From                Message
  ----    ------                ----  ----                -------
  Normal  EnsuringLoadBalancer  24m   service-controller  Ensuring load balancer
  Normal  EnsuredLoadBalancer   24m   service-controller  Ensured load balancer
  Normal  EnsuringLoadBalancer  19m   service-controller  Ensuring load balancer
  Normal  EnsuredLoadBalancer   19m   service-controller  Ensured load balancer
```

```
ubuntu@ip-172-31-46-170:~$ kubectl delete svc nginx-deployment
service "nginx-deployment" deleted from default namespace
ubuntu@ip-172-31-46-170:~$ kubectl expose deployment nginx-deployment \
  --name nginx-service \
  --type=LoadBalancer \
  --port=80 \
  --target-port=80
Error from server (AlreadyExists): services "nginx-service" already exists
ubuntu@ip-172-31-46-170:~$ kubectl delete svc nginx-service
service "nginx-service" deleted from default namespace
ubuntu@ip-172-31-46-170:~$ kubectl expose deployment nginx-deployment \
  --name nginx-service \
  --type=LoadBalancer \
  --port=80 \
  --target-port=80
service/nginx-service exposed
ubuntu@ip-172-31-46-170:~$ kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME             ENDPOINTS                                   AGE
nginx-service   192.168.34.226:80,192.168.57.32:80   25s
ubuntu@ip-172-31-46-170:~$ kubectl get svc nginx-service
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
nginx-service   LoadBalancer   10.100.109.126   a7f1facd6fe40446482567794ab918fd-1428899793.eu-north-1.elb.amazonaws.com   80:32049/TCP   44s
ubuntu@ip-172-31-46-170:~$ kubectl get endpoints nginx-service
kubectl get svc nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME             ENDPOINTS                                   AGE
nginx-service   192.168.34.226:80,192.168.57.32:80   2m40s
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
nginx-service   LoadBalancer   10.100.109.126   a7f1facd6fe40446482567794ab918fd-1428899793.eu-north-1.elb.amazonaws.com   80:32049/TCP   2m40s
ubuntu@ip-172-31-46-170:~$ kubectl get pods -l app=nginx
```

```
ubuntu@ip-172-31-46-170:~$ kubectl get pods -l app=nginx
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75fdcbbc74-6qtg9   1/1     Running   0          22m
nginx-deployment-75fdcbbc74-75bwk   1/1     Running   0          22m
ubuntu@ip-172-31-46-170:~$ kubectl describe pod <pod-name>
-bash: syntax error near unexpected token `newline'
ubuntu@ip-172-31-46-170:~$ kubectl describe pod nginx-deployment-75fdcbbc74-6qtg9
Name:             nginx-deployment-75fdcbbc74-6qtg9
Namespace:        default
Priority:         0
Service Account:  default
Node:             ip-192-168-43-111.eu-north-1.compute.internal/192.168.43.111
Start Time:       Tue, 17 Feb 2026 09:40:40 +0000
Labels:           app=nginx
                  pod-template-hash=75fdcbbc74
                  topology.kubernetes.io/region=eu-north-1
                  topology.kubernetes.io/zone=eu-north-1c
Annotations:      <none>
Status:           Running
IP:               192.168.34.226
IPs:
  IP:           192.168.34.226
Controlled By:  ReplicaSet/nginx-deployment-75fdcbbc74
Containers:
  nginx:
    Container ID:   containerd://62aaea2a457bbe4ecdd7749d70167ed3e034fab7ee25e422f853259d3940567c
    Image:          nginx
    Image ID:       docker.io/library/nginx@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Tue, 17 Feb 2026 09:40:41 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-rqnjp (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-rqnjp:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
```

```
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  23m   default-scheduler  Successfully assigned default/nginx-deployment-75fdcbbc74-6qtg9 to ip-192-168-43-111.eu-north-1.compute.internal
  Normal  Pulling    23m   kubelet            spec.containers{nginx}: Pulling image "nginx"
  Normal  Pulled     23m   kubelet            spec.containers{nginx}: Successfully pulled image "nginx" in 710ms (710ms including waiting). Image size: 62939286 bytes.
  Normal  Created    23m   kubelet            spec.containers{nginx}: Created container: nginx
  Normal  Started    23m   kubelet            spec.containers{nginx}: Started container nginx
ubuntu@ip-172-31-46-170:~$ kubectl get deployment nginx-deployment
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           24m
ubuntu@ip-172-31-46-170:~$ kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
ubuntu@ip-172-31-46-170:~$ curl http://a7f1facd6fe40446482567794ab918fd-1428899793.eu-north-1.elb.amazonaws.com
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@ip-172-31-46-170:~$ kubectl get svc
NAME            TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1       <none>                                                                   443/TCP        53m
nginx-service   LoadBalancer   10.100.109.126   a7f1facd6fe40446482567794ab918fd-1428899793.eu-north-1.elb.amazonaws.com   80:32049/TCP   9m33s
ubuntu@ip-172-31-46-170:~$
```

9. Open this in browser